

Lab: When Your Hands Think Ahead

Materials Exploration — Cognition

Objective

Explore the cognitive dimensions of working with yarn — flow states, material prediction, counterfactual thinking, and the relationship between challenge and skill. You will develop awareness of your own thinking-through-making processes.

Materials Needed

- Your most familiar yarn (a fiber and weight you know well)
- An unfamiliar yarn (a fiber or weight you rarely use)
- A crochet hook appropriate for each
- A pattern with moderate complexity (e.g., a granny square or textured stitch pattern)
- A timer
- A notebook for observations

Part 1: Mapping Your Flow Zone (20 minutes)

1. **Easy task with familiar yarn:** Crochet a simple chain stitch for 5 minutes with your most familiar yarn. Note:
 2. How much conscious attention does this require? (1-5 scale)
 3. Does your mind wander? What does it wander to?
 4. Is this flow, or is it too easy for flow?
5. **Moderate task with familiar yarn:** Crochet a textured stitch pattern (e.g., moss stitch, shell stitch) for 5 minutes with the same yarn. Note:
 6. How much conscious attention does this require?
 7. Do you feel absorbed? Is this closer to flow?

8. **Hard task with unfamiliar yarn:** Attempt the same textured pattern with the unfamiliar yarn. Note:

9. How much conscious attention does this require?

10. Is this engaging-but-manageable, or frustrating?

11. How is the experience different from the familiar yarn?

Record your findings:

Condition	Attention (1-5)	Emotional Tone	Flow? (Y/N)	Notes

Reflection: Where was your "flow zone"? What combination of challenge and material familiarity produced the most absorbing experience?

Part 2: Testing Your Generative Model (15 minutes)

1. Without looking at the label, pick up your familiar yarn and write down everything you predict about its behavior:

2. Stitch definition (crisp or soft?)

3. Splitting tendency

4. Drape of finished fabric

5. How it will respond to washing

6. How it will feel after several hours of handling

7. Now do the same for the unfamiliar yarn. Write down your best predictions.

8. Crochet a 10-row swatch with each yarn. Compare your predictions to reality.

Reflection: How much richer is your generative model for the familiar yarn? Were any of your predictions for the unfamiliar yarn wrong? What did you learn?

Part 3: Counterfactual Thinking Exercise (15 minutes)

1. Choose a simple pattern (like a scarf or dishcloth).

2. The pattern calls for worsted-weight cotton. Without crocheting a stitch, mentally simulate what would happen if you used:
3. Fingering-weight silk
4. Bulky acrylic
5. Lace-weight mohair
6. For each substitution, predict:
7. How would gauge change?
8. How would drape change?
9. How would the visual texture change?
10. Would the item still be functional for its purpose?
11. If you have samples of these yarns, crochet a few rows with each to test your predictions.

Reflection: How accurate was your mental simulation? Counterfactual thinking is your generative model running "offline" — generating predictions without sensory feedback. How confident did you feel in your simulations?

Part 4: The Frogging Experiment (10 minutes)

1. Crochet 10 rows of any stitch pattern.
2. Intentionally make a mistake in row 5 (skip a stitch, use the wrong stitch, change tension dramatically).
3. Continue crocheting past the mistake for 5 more rows.
4. At what point did you notice the mistake? Was it immediate or delayed?
5. Decide: frog back to fix it, or live with it?
6. If you frog, notice the cognitive experience of undoing work. Does it feel like relief, frustration, or something else?

Reflection: Frogging is a cognitive reset — acknowledging that the model and the fabric have diverged. How did you feel about the decision? What does your tolerance for imperfection tell you about your own free energy threshold?

Discussion Questions

1. Was your flow experience in Part 1 what you expected? Did the unfamiliar yarn disrupt flow, or did it create a different kind of engagement?
2. How did counterfactual thinking compare to actually crocheting with different yarns? Was your mental model accurate?
3. What did the frogging exercise reveal about your relationship to mistakes and prediction error?

Reflection

Write a short paragraph about thinking and making. How does crochet combine mental and manual cognition? What did this lab teach you about the intelligence embedded in your hands and your relationship with materials?